

# SIMATS ENGINEERING

## ASSESSMENT TOOL 1

**COURSE CODE: CSA14**

**COURSE NAME: Compiler Design**

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### COURSE OUTCOME COVERED

**CO4:** *Examine intermediate code generation techniques to enhance code optimization (BL4)*

*Assessment Tool Weightage: 40%*

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**QUESTIONS: 4**

**Total Marks:50**

Annexure A

### CLASS TEST

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#### ***Code of Conduct:***

*I \_\_\_\_K. Deepthi(192525343)\_\_\_\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student: K. Deepthi***

| Q. No                        | Understanding of Concepts (25) | Explanation (25) | Application (20) | Organization (15) | Accuracy(10) | Total Score (out of 100) |
|------------------------------|--------------------------------|------------------|------------------|-------------------|--------------|--------------------------|
| 1                            | / 4                            | / 4              | / 4              | / 4               | / 4          |                          |
| 2                            | / 4                            | / 4              | / 4              | / 4               | / 4          |                          |
| 3                            | / 4                            | / 4              | / 4              | / 4               | / 4          |                          |
| 4                            | / 4                            | / 4              | / 4              | / 4               | / 4          |                          |
| 5                            | / 4                            | / 4              | / 4              | / 4               | / 4          |                          |
| <b>OVERALL RUBRICS VALUE</b> |                                |                  |                  |                   |              |                          |
|                              | / 4                            | / 4              | / 4              | / 4               | / 4          |                          |
|                              | <b>/ 25</b>                    | <b>/ 30</b>      | <b>/ 20</b>      | <b>/ 15</b>       | <b>/ 10</b>  | <b>/ 100</b>             |

## Annexure A

### CLASS TEST

Q1. Construct three address code for the following

if((a < b) and ((c < d) or (a > d))) then

z=x + y \*z

else

z = z + 1

Q2.Translate (a < b) and (c < d) or (e < f) into three address statements using back patching.

Q3. Construct three address code using the following production for Boolean expression (a > b) or (c < d) or (e < f)

$B \rightarrow B1 \parallel B2$

$B \rightarrow B1 \ \&\& \ B2$

$B \rightarrow ! \ B1$

$B \rightarrow E1 \ \text{relop} \ E2$

$B \rightarrow \text{true}$

$B \rightarrow \text{false}$

Q4. Generate the three address code for the following program fragment-

while (A < C and B > D) do

if A = 1 then C = C + 1

else while A < = D do

A = A + B

18/08/26

CO4 - AT1

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Course code: CSA1402

1. Construct three Address code for the following if (a < b) and ((c < d) or (a > d)) then  $z = x + y$  else  $z = z + 1$

Given statement:

if (a < b) and ((c < d) or (a > d)) then  $z = z + 1$

Step 1: understand the boolean conditions

(a < b) AND ((c < d) OR (a > d))

a < b  
c < d  
a > d

The structure is (a < b) AND [(c < d) OR (a > d)]

Step 2: Create Tables

use labels for the conditions checks, then part, else part, and the end of the statement.

L<sub>1</sub>, L<sub>2</sub>, L<sub>3</sub>, L<sub>4</sub>, L<sub>5</sub>, L<sub>6</sub>

Step 3: Generate TAC for the condition

L<sub>1</sub>:

if a < b goto L<sub>2</sub>  
goto L<sub>5</sub>

L<sub>2</sub>:

if c < d goto L<sub>3</sub>  
goto L<sub>5</sub>

L3:  
if  $a > d$  goto L4  
goto L5

If  $a < b$  is false, the complete AND expression is false.  
So control goes to the else part. If  $a < b$  is true, the expression is evaluated. If either  $c < d$  or  $a > d$  is true, control goes to the then part.

Step 4:

Generate TAC for the then part the statement is  $z = x + y * z$ . Multiplication has higher precedence.

L4:

$t_1 = y * z$

$t_2 = x + t_1$

$z = t_2$

goto L6

Step 5: Generate TAC for the ELSE Part

L5:

$t_3 = z + 1$

$z = t_3$

Final TAC

L1:

if  $a < b$  goto L2

goto L5

L2:

if  $c < d$  goto L4

goto L3

L3:

if  $a > d$  goto L4

goto L5

L4:

$t_1 = y * z$

$t_2 = x + t_1$

$z = t_2$

goto L6

L5:

$$t_3 = z + 1$$

$$z = t_3$$

2) Translate  $(a < b)$  and  $(c < d)$  or  $(e < f)$  into three address statements using back patching.

Given,  $(a < b)$  and  $(c < d)$  or  $(e < f)$

Step-1: Apply operator precedence  
AND has higher precedence than OR, so the expression is  $[(a < b) \text{ AND } (c < d)] \text{ OR } (e < f)$

Step-2: Generate incomplete jumps

1. if  $a < b$  goto -

g. goto -

3. if  $c < d$  goto -

u. goto -

5. if  $e < f$  goto -

6. goto -

The underscore represents a target address

Step-3: Handle AND

For  $(a < b) \text{ AND } (c < d)$ , if  $a < b$  is true, control must continue to the 2nd condition.

Backpatch statement 1 to statement 3.

1. if a < b goto 3

Step 4:-

Handle OR  
If the first AND expression is false, the second OR operand ecf must be evaluated.

Back Patch statement 2 to statement 5

Back Patch statement 4 to statement 5

Step 5:- Final back patched code.

1. if a < b goto 3

2. goto 5

3. if c < d goto 7

4. goto 5

5. if e < f goto 7

6. goto 8

7. TRUE

8. FALSE

Backpatching means generating jump instructions  
1st and filling target address.

3) Construct three address code using the following Production  
for Boolean expression  $(a > b)$  or  $(c < d)$  or  $(e < f)$

$B \rightarrow B_1 \parallel B_2$

$B \rightarrow B_1 \&\& B_2$

$B \rightarrow !B_1$

$B \rightarrow E_1 \text{ relop } E_2$

$B \rightarrow \text{true}$

$B \rightarrow \text{false}$

L1: if  $(a > b)$  goto L3  
goto L2

Step 2: L2:  
if  $(a > b)$  goto LTRUE  
goto L2

Step 3: L3:  
if  $c < d$  goto LTRUE  
goto L3

Step 4: L3:  
if  $c < d$  goto LTRUE  
if  $c < d$  goto LTRUE  
goto LFALSE

Final TAC for Q9

L1:  
if  $a > b$  goto LTRUE  
goto L2

- 4) Generate the three address code for the following program fragment while  $(A < C \text{ and } B > D)$  do if  $A = 1$  then  $C = C + 1$  else while  $A < = D$  do  $A = A + B$

Complete TAC

L1: if  $A < C$  goto L2  
goto LEND

L2: if  $B > D$  goto L3  
goto LEND

L3: if  $A == 1$  goto L4  
goto L5

L4:  $t_1 = C + 1$   
 $C = t_1$   
goto L1

L5: if  $A < D$  goto L6  
goto L1

L6:  
 $t_2 = A + B$   
 $A = t_2$   
goto L5

## RUBRICS FOR CALCULATION

| Criteria                  | Weightage | Level 4 – Exemplary                                         | Level 3 – Proficient                    | Level 2 – Developing                         | Level 1 – Beginning                      |
|---------------------------|-----------|-------------------------------------------------------------|-----------------------------------------|----------------------------------------------|------------------------------------------|
| Understanding of Concepts | 25%       | Demonstrates thorough, accurate understanding of concepts   | Good understanding with minor gaps      | Basic understanding with some misconceptions | Poor understanding; major misconceptions |
| Explanation               | 25%       | Detailed, well-explained answers with relevant examples     | Clear explanation with adequate details | Limited explanation; lacks depth             | Poor or unclear explanation              |
| Application               | 20%       | Effectively applies concepts with strong analytical ability | Applies concepts with minor errors      | Limited application with weak analysis       | No application or incorrect analysis     |
| Organization              | 15%       | Well-structured answers with logical flow and coherence     | Organized with minor issues in flow     | Basic structure, lacks clarity               | Poor organization, incoherent answers    |
| Accuracy                  | 15%       | Completely accurate and covers all required points          | Mostly accurate with minor omissions    | Partially correct with missing points        | Incorrect answers with major gaps        |